

Background and rationale

- Kelp forests cover over a third of global coastal systems and play a key role in marine biodiversity and carbon sequestration.
- Sugar kelp (*Saccharina latissima*) is a commercially important species across Ireland, Northern Ireland and Scotland, with growing applications in aquaculture, biofuel and biotechnology.
- Understanding genetic structure is fundamental to kelp conservation and sustainable management, as it identifies genetically distinct populations requiring protection and guides sourcing of locally adapted strains for cultivation.
- Current neutral microsatellite resources for *S. latissima* are insufficient for fine-scale population genetic studies, limiting statistical power for detecting weak population structure

Aim

To develop and validate a panel of polymorphic microsatellite markers for *S. latissima* to enable high-resolution population genetic studies

Study objectives

Scan *S. latissima* reference genome (GCA_034768055.1) for tri-, tetra- and pentanucleotide microsatellites using Krait2

Design primer pairs with Primer3 and validate specificity by BLASTn against the reference genome (Galaxy platform)

Validate amplification by gel electrophoresis PCR across 3 geographically distant populations

Confirm polymorphism via capillary electrophoresis (ABI Analyser 3730xl)

Develop multiplex panels from polymorphic loci

Study design & sampling

682

individuals sampled

17

sites (NI, ROI, Scotland)

Collection period: August 2024 – July 2025. A 1 cm² disc from the meristematic region of each blade stored in silica gel or RNAlater at –20°C.

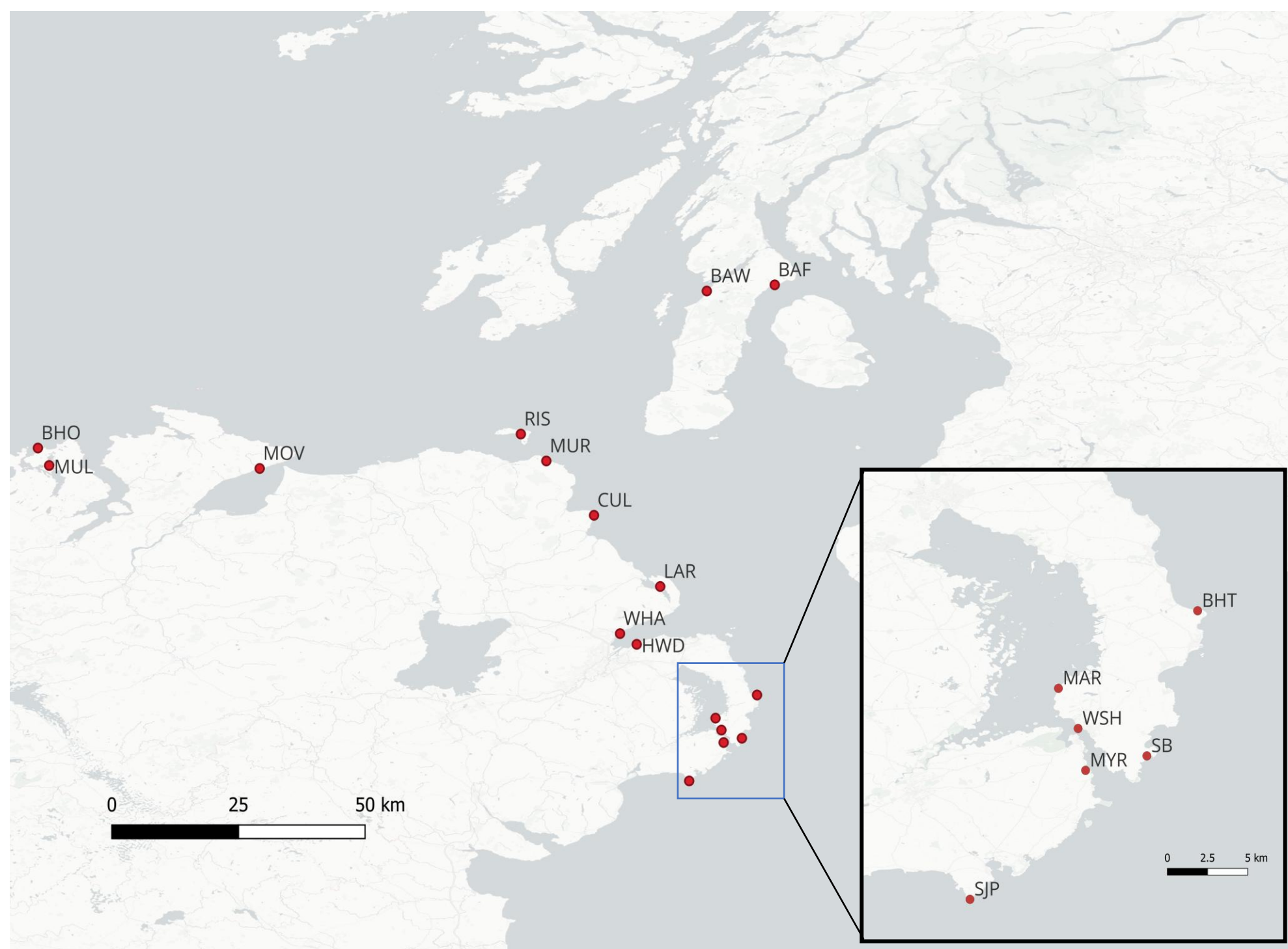


Figure 1. Map showing the location of sample sites

In silico pipeline

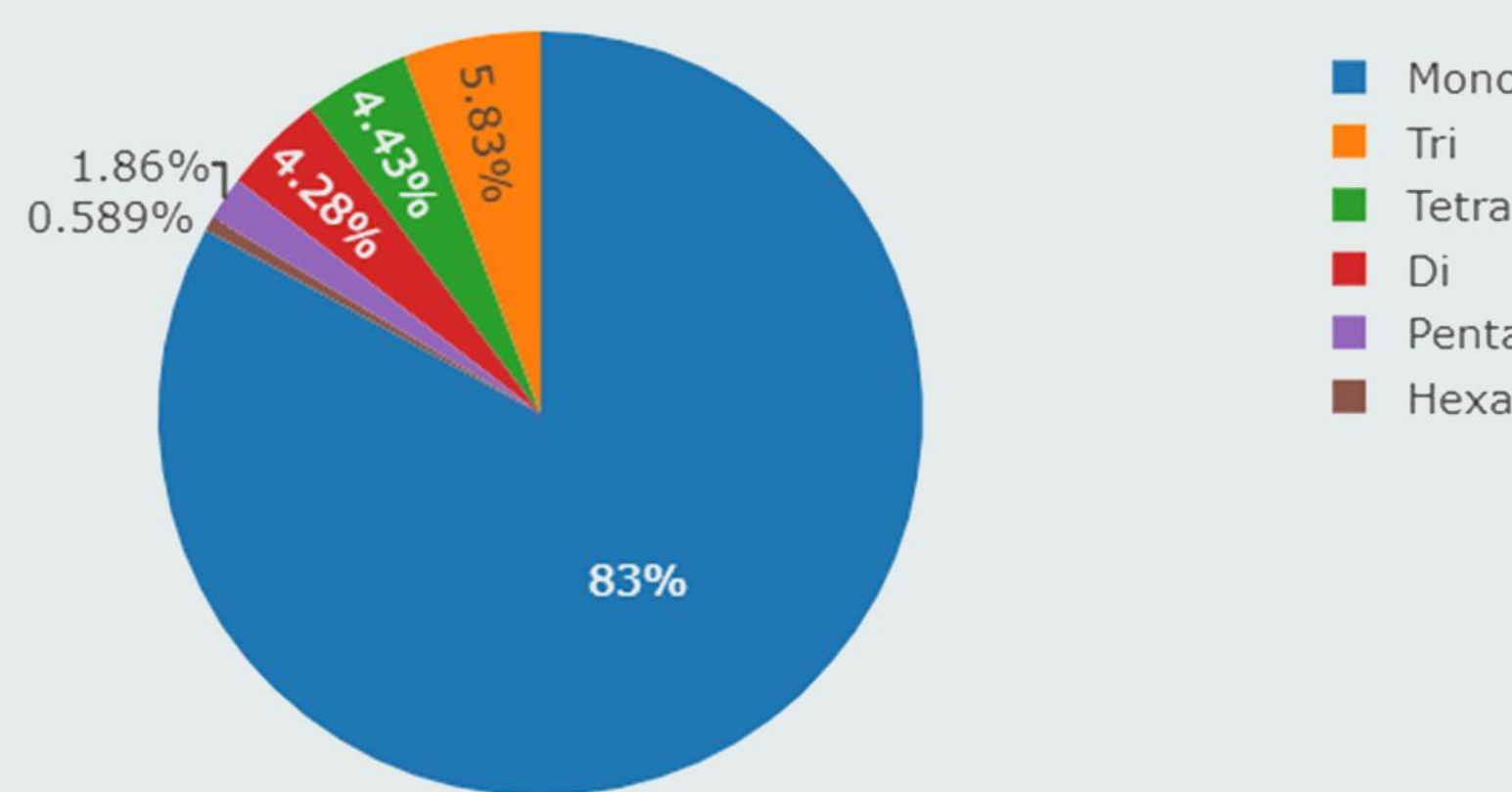
Step 1 — Microsatellite detection: Krait2

Genome scanned for perfect microsatellites using Krait2 (Du et al. 2025). Only tri-, tetra- and pentanucleotide motifs targeted for marker development

Motif	Min Repeats	Included?
Mono	--	No
Di	--	No
Tri	≥8	Yes
Tetra	≥7	Yes
Penta	≥7	Yes

Total microsatellites: **1,308,173**
Total microsatellites after filtering: **5,310**

SSR counts distribution for each type



Step 2 — Primer design: Primer3

Primer3 integrated within Krait2 used to design flanking primers for all retained microsatellite loci. Parameters optimised for kelp amplification:

Primer length	Annealing temp
18–25 bp (optimum 22 bp)	52 – 60°C (optimum 58°C)
GC content	Flanking sequence
30 – 60% (optimum 40%)	100 bp
3' end stability	Product size
> 4 (primer-dimer minimisation)	100 – 250

Step 3 — Specificity: BLASTn via Galaxy

All primer pairs aligned against the full *S. latissima* assembly (GCA_034768055.1) using BLASTn within the Galaxy platform (Afgan et al. 2018) to confirm single-locus specificity.

Tool	BLASTn (nucleotide)
Platform	Galaxy
E-cutoff value	0.001
Word size	7
Exclusion criteria	Multiple high-confidence or off-target hits
Retained	Uniquely mapping primers

Outcome: 104 primer pairs retained from 5,310 filtered microsatellite loci

In silico pipeline cont

Step 4 — PCR validation & capillary electrophoresis

44 primer pairs selected for empirical testing across 3 geographically distant *S. latissima* populations.

PCR reaction

Component	Concentration	Volume (μL)
PPP mix	2×	1.75
Standard primer	10 μM	0.1
Fluorescent primer	10 μM	0.1
DMSO	100%	0.175
BSA	0.2 μg/μL	0.3
Template DNA	-	1
H ₂ O	-	0.075
Total		3.5

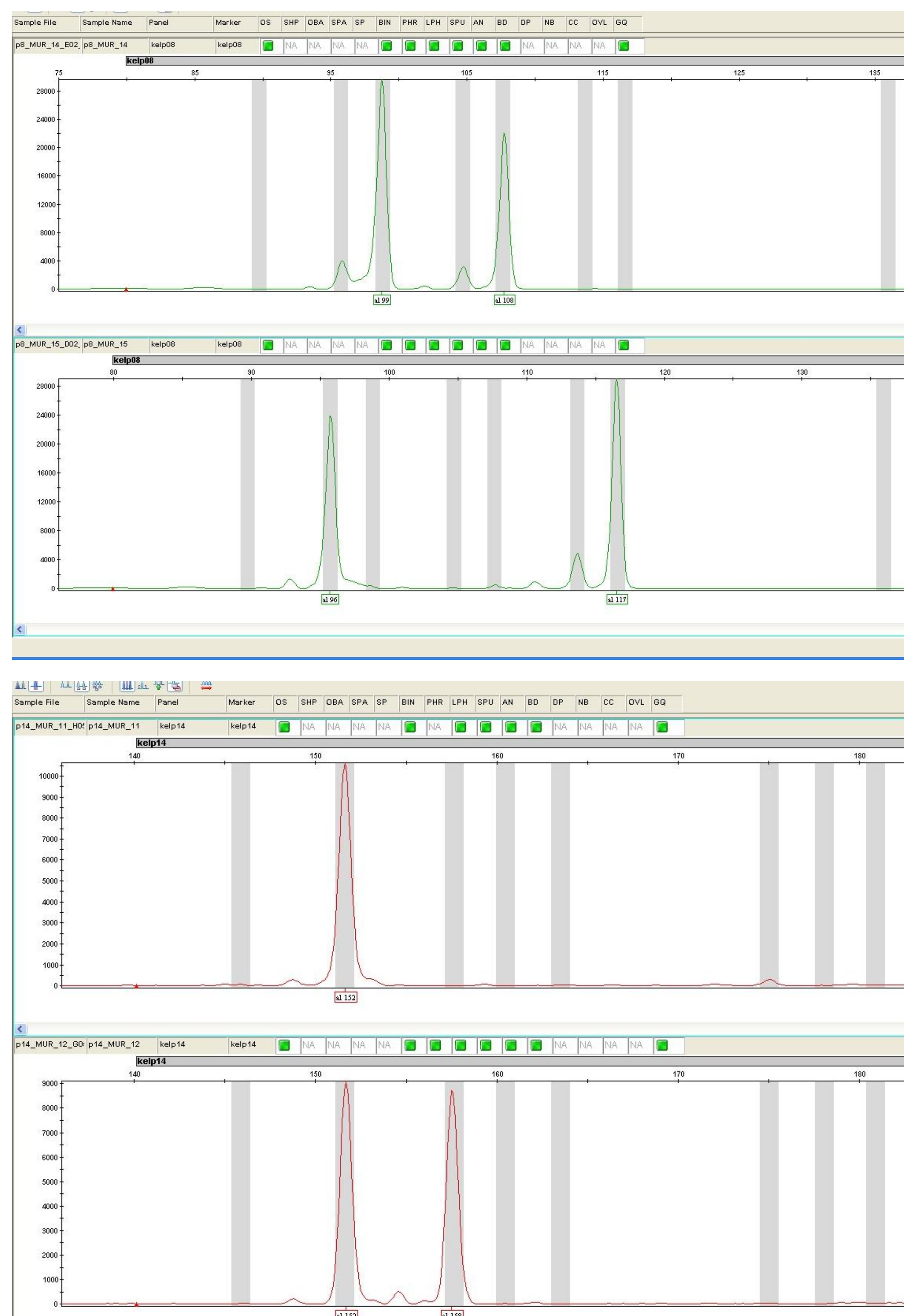


Figure 3. Output of microsatellite fragment analysis and allele binning

Filtering & validation outcomes

Microsatellites detected (Krait2)	1,308,173
Motifs targeted	Tri, tetra, penta only
Primer pairs after BLASTn	104
Selected for PCR testing	44
Successfully amplified	30 (68%)
Confirmed polymorphic	20
Novel loci in final panel	13
Published loci (Paulino et al. 2016)	7
Total panel size	20 loci
Multiplex panels designed	5